

torch.polar#

<https://pytorch.org/docs/stable/generated/torch.polar.html#torch.polar>

Description:

Constructs a complex tensor whose elements are Cartesian coordinates corresponding to the polar coordinates with absolute value `abs` and angle `angle`. `torch.polar` is similar to `std::polar` and does not compute the polar decomposition of a complex tensor like Python's `cmath.polar` and SciPy's `linalg.polar` do. The behavior of this function is undefined if `abs` is negative or NaN, or if `angle` is infinite. `abs` (Tensor) – The absolute value of the complex tensor. Must be float or double. `angle` (Tensor) – The angle of the complex tensor. Must be same dtype as `abs`.

Function Signature

```
torch.polar(abs, angle, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor) – If the inputs are torch.float32, must be torch.complex64. If the inputs are torch.float64, must be torch.complex128.

Code Examples

```
>>> import numpy as np
>>> abs = torch.tensor([1, 2], dtype=torch.float64)
>>> angle = torch.tensor([np.pi / 2, 5 * np.pi / 4],
dtype=torch.float64)
>>> z = torch.polar(abs, angle)
>>> z
tensor([(0.0000+1.0000j), (-1.4142-1.4142j)],
dtype=torch.complex128)
```

Notes & Warnings

NOTE Note torch.polar is similar to std::polar and does not compute the polar decomposition of a complex tensor like Python's cmath.polar and SciPy's linalg.polar do. The behavior of this function is undefined if abs is negative or NaN, or if angle is infinite.

Detailed Documentation

Documentation

Constructs a complex tensor whose elements are Cartesian coordinates corresponding to the polar coordinates with absolute value `abs` and angle `angle`.

`torch.polar` is similar to `std::polar` and does not compute the polar decomposition of a complex tensor like Python's `cmath.polar` and SciPy's `linalg.polar` do. The behavior of this function is undefined if `abs` is negative or NaN, or if `angle` is infinite.

`abs` (Tensor) – The absolute value the complex tensor. Must be float or double.

`angle` (Tensor) – The angle of the complex tensor. Must be same dtype as `abs`.

`out` (Tensor) – If the inputs are `torch.float32`, must be `torch.complex64`. If the inputs are `torch.float64`, must be `torch.complex128`.